Project Documentation

Title

Smart Bus & Road Safety Monitoring System with LoRa-Based Communication

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Abstract

This project integrates **ESP32 microcontroller, GSM/GPS modules, ADXL345 accelerometer, ultrasonic sensors, and LoRa SX1278 transceivers** to build a comprehensive bus safety monitoring system. The solution detects multiple accident scenarios, monitors passenger count, and provides real-time alerts via LCD, buzzer, GSM calls/SMS, and LoRa communication. The LoRa module enables long-range wireless signaling between vehicles, ensuring idle buses can detect nearby trucks and trigger safety alerts.

Objectives

- Monitor passenger overcrowding and disable engine when unsafe.
- Detect driver alcohol consumption, smoke, flame, and rain conditions.
- Provide accident detection using ADXL345 accelerometer.
- Implement collision guard using ultrasonic sensor + servo.
- Enable GSM/GPS alerts with live location via SMS/calls.
- Integrate LoRa communication for inter-vehicle safety signaling.
- Display real-time status on LCD for user feedback.

Hardware Components

- **ESP32 microcontroller** (main controller)
- **ATmega328 (Arduino Uno/Nano)** for LoRa prototype
- **LoRa SX1278 transceiver** (433 MHz)
- **ADXL345 accelerometer** (impact detection)
- **Ultrasonic sensor (HC-SR04)** for collision guard
- **IR sensors** for passenger entry/exit detection
- **Alcohol sensor (MQ-3)**
- **Flame sensor, Smoke sensor**
- **Rain sensor**
- **Relay modules** (engine, CO_2 sprinkler, glass breaker)
- **LCD (20x4 I²C)** for real-time display
- **GSM (SIM800) & GPS modules** for communication and tracking
- **Buzzer + Servo motor** for alerts

Hardware Components with Pin & Voltage Details

Component	Operating Voltage	Key Pins	Integration Notes
ESP32 Microcontroller	3.3V (5V via USB/VIN)	GPIO pins, 3.3V, GND	Central controller; use level shifters for 5V sensors.
ATmega328 (Arduino)	5V	Digital D0–D13, Analog A0–A5	Used for LoRa prototype interfacing.
LoRa SX1278 (433 MHz)	3.3V	MOSI, MISO, SCK, NSS (SPI)	Requires 3.3V regulator; connect via SPI.
ADXL345 Accelerometer	3.3V	SDA, SCL (I^2C) or SPI pins	Detects impacts; typically uses I^2C with ESP32.
Ultrasonic (HC-SR04)	5V	VCC, GND, TRIG, ECHO	**Warning:** ECHO pin needs a voltage divider for ESP32.
IR Sensors	5V	VCC, GND, OUT	Used for passenger entry/exit detection.
Alcohol Sensor (MQ-3)	5V	VCC, GND, AO, DO	AO connects to ESP32 ADC for concentration.
Flame Sensor	3.3–5V	VCC, GND, DO	Detects fire presence.
Gas Sensor (MQ-2)	5V	VCC, GND, AO, DO	Detects smoke/LPG. Needs preheating for stability.
Rain Sensor	5V	VCC, GND, AO, DO	AO connects to ADC for intensity classification.
Relay Modules	5V control	VCC, GND, IN	Controls engine cutoff and safety systems.
LCD (20x4 I^2C)	5V (3.3V Logic)	SDA, SCL	Displays passenger count, alerts, and speed.
GSM (SIM800) Module	3.7–4.2V	TX, RX, VCC, GND	Use Li-ion battery; ensure UART level shifting.
GPS (NEO-6M)	3.3–5V	TX, RX, VCC, GND	Provides live location data via UART.
Buzzer	3.3–5V	VCC, GND, IN	Provides audible alerts.

Component	Operating Voltage	Key Pins	Integration Notes
Servo Motor	5V	VCC, GND, PWM signal	Activates physical collision guards.

Software Tools

- **Arduino IDE** (programming ESP32 & ATmega328)
- **Proteus** (simulation of circuits)
- **C/Embedded C** (firmware development)

System Architecture

1. **Input Layer**: Sensors (IR, alcohol, flame, smoke, rain, ultrasonic, accelerometer).
2. **Processing Layer**: ESP32 microcontroller integrates sensor data, applies safety logic, and triggers events.
3. **Output Layer**: LCD display, buzzer, relays, GSM/GPS alerts, LoRa communication.
4. **Communication Layer**:
   - GSM/GPS → sends SMS/calls with live location.
   - LoRa → transmits safety signals between vehicles (e.g., idle bus detects nearby truck).

Working Principle

The Smart Bus & Road Safety Monitoring System operates as a continuous monitoring and control framework, integrating multiple sensors, actuators, and communication modules. The ESP32 microcontroller acts as the central hub, processing sensor inputs and executing safety logic in real time.

1. **Passenger Monitoring**
   - IR sensors at the bus door detect entry and exit events.
   - The system maintains a live passenger count, displayed on the LCD.
   - If overcrowding is detected (count > 5), the engine relay is disabled, a buzzer alert is triggered, and a GSM SMS is sent to the control center.
   - When passenger count returns to safe levels, the engine relay is re-enabled.
2. **Driver Safety (Alcohol Detection)**
   - The MQ-3 alcohol sensor continuously monitors the driver’s breath.
   - If alcohol concentration exceeds the threshold, the system gradually reduces engine speed using PWM control, preventing sudden stops.
   - A GSM call and SMS are sent to authorities with GPS location, and the LCD displays “ALCOHOL ALERT.”
3. **Fire and Smoke Detection**
   - Flame and smoke sensors monitor the cabin environment.
   - On detection, the buzzer sounds, the CO_2 sprinkler relay activates, and the glass breaker relay is triggered for emergency evacuation.
   - GSM alerts are sent, and the LCD shows “FIRE ALERT” or “SMOKE ALERT” with sensor values.

4. **Rain-Based Speed Control**
    - The rain sensor measures intensity and classifies conditions into low, moderate, or heavy rain.
    - Depending on rain severity, the system automatically reduces bus speed to safe limits (70, 65, or 60 km/h thresholds).
    - Alerts are displayed on the LCD and sent via GSM to notify the driver and control center.
5. **Collision Guard (Ultrasonic + Servo)**
    - The ultrasonic sensor measures distance to vehicles ahead.
    - If another vehicle is detected within unsafe range, the servo guard is activated, buzzer sounds, and LCD shows "GUARD ALERT."
    - Different thresholds are applied for low-speed (2 m) and high-speed (7 m) scenarios.
6. **Accident Detection (ADXL345 Accelerometer)**
    - The accelerometer measures total acceleration across axes.
    - If sudden impact exceeds the threshold (12 g), the system identifies an accident.
    - Engine relay is cut off, buzzer sounds continuously, and GSM sends accident alerts with GPS coordinates.
7. **Communication Layer**
    - **GSM/GPS**: Sends SMS/calls with live location to authorities during alcohol, fire, smoke, rain, or accident events.
    - **LoRa**: Provides long-range communication between buses and trucks. For example, an idle bus detects a nearby truck via LoRa and triggers LED alerts, enhancing roadside safety.
8. **User Feedback (LCD + Buzzer)**
    - The LCD continuously displays passenger count, speed, and active alerts.
    - The buzzer provides immediate audible warnings for critical events.
    - Together, they ensure both driver and passengers are aware of the bus's safety status.

Results

- Successfully integrated **multi-sensor accident detection** with ESP32.
- Real-time alerts sent via **LCD, buzzer, GSM SMS/calls**.
- LoRa communication enabled **long-range inter-vehicle signaling**.
- System demonstrated improved **road safety and passenger monitoring**.

Applications

- Public transport fleets (buses, school vans).
- Smart city infrastructure for accident prevention.
- Parking and idle vehicle monitoring using LoRa.

Conclusion

The project demonstrates a **multi-scenario bus safety monitoring system** combining IoT, embedded systems, and wireless communication. By integrating GSM/GPS with LoRa, the solution ensures both **local safety monitoring** and **long-range inter-vehicle alerts**, making it suitable for smart city and fleet management applications.

Future Scope

- Cloud integration for centralized monitoring.
- AI-based predictive accident detection.
- Expansion to multiple vehicles with mesh LoRa networks.
- Integration with government transport safety dashboards.

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YARRADODDI NIKHIL KUMAR REDDY

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MADHU

SOWMYA

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